

Marking SCHEME OF CAT N⁰2: Calculus for Engineers /20 Marks

Question1.

a) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined as $T(\vec{x}) = \langle x_2, x_1, x_1 + x_2 \rangle$

T is a linear operator iff $\forall \vec{x}, \vec{y} \in \mathbb{R}^2, \exists \alpha, \beta \in \mathbb{R}$ such that $T(\alpha\vec{x} + \beta\vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$ /1 Mark

$\vec{x} = \langle x_1, x_2 \rangle$ and $\vec{y} = \langle y_1, y_2 \rangle$

Verification:

$T(\alpha\vec{x} + \beta\vec{y}) = \langle \alpha x_2 + \beta y_2, \alpha x_1 + \beta y_1, \alpha x_1 + \beta y_1 + \alpha x_2 + \beta y_2 \rangle$ /1 Mark

$\alpha T(\vec{x}) + \beta T(\vec{y}) = \alpha \langle x_2, x_1, x_1 + x_2 \rangle + \beta \langle y_2, y_1, y_1 + y_2 \rangle$ /1 Mark

As $T(\alpha\vec{x} + \beta\vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$, then we conclude that T is a linear operator /1 Mark

b) $u_k = \left(\frac{2k-5}{4k+1}\right)^k \Rightarrow \sqrt[k]{\left(\frac{2k-5}{4k+1}\right)^k} \Rightarrow \lim_{k \rightarrow \infty} \sqrt[k]{\left(\frac{2k-5}{4k+1}\right)^k} = \frac{1}{2} < 1$ /2 Marks

Hence,

$\sum_{k=1}^{+\infty} \left(\frac{2k-5}{4k+1}\right)^k$ Converged /1 Mark

Question2

a) An 2×2 matrix Q is **orthogonal matrix** if and only if $Q^T Q = I$ / 0.5Mark

$Q^T Q = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$ / 1Mark

Hence, $Q = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ is an orthogonal / 0.5Mark

b) Since $|(-1)^k| = |(-1)^{k+1}| = 1$, we obtain

$q = \lim_{k \rightarrow \infty} \frac{|u_{k+1}|}{|u_k|} = \lim_{k \rightarrow \infty} \left| \frac{x^{k+1}}{3^{k+1}(k+2)} \cdot \frac{3^k(k+1)}{x^k} \right| = \lim_{k \rightarrow \infty} \left| \frac{|x|}{3} \cdot \frac{(k+1)}{(k+2)} \right| = \frac{|x|}{3}$ / 1Mark

The series converges absolutely if $q < 1 \Leftrightarrow \frac{|x|}{3} < 1 \Leftrightarrow |x| < 3$, and diverges if $|x| > 3$ / 1Mark

The ratio test fails when $|x| = 3$, so the case $x = 3$ and $x = -3$ need separate analyse.

Substituting $x = -3$ in a given series yield:

$$\sum_{k=0}^{+\infty} \frac{(-1)^k (-3)^k}{3^k (k+1)} = \sum_{k=0}^{+\infty} \frac{3^k}{3^k (k+1)} = \sum_{k=0}^{+\infty} \frac{1}{(k+1)} \quad / \text{ 0.5Mark}$$

Which is the divergent harmonic series.

Substituting $x = 3$ in a given series yield:

$$\sum_{k=0}^{+\infty} \frac{(-1)^k 3^k}{3^k (k+1)} = \sum_{k=0}^{+\infty} \frac{(-1)^k}{(k+1)} \quad / \text{ 0.5Mark}$$

Which is conditionally convergent alternating harmonic series.

Thus the interval of convergence for the given series is $(-3, 3]$ and the radius of convergence is $R = 3$. / **2Marks**

Question3.

$$\det(A - \lambda I) = 0 \Leftrightarrow \begin{vmatrix} 2-\lambda & -3 & 1 \\ 1 & -2-\lambda & 1 \\ 1 & -3 & 2-\lambda \end{vmatrix} = 0 \Leftrightarrow -\lambda(\lambda-1)^2 = 0 \rightarrow \lambda_1 = 0, \lambda_2 = \lambda_3 = 1 \quad \text{1Mark}$$

$$\text{For } \lambda = 0, \text{ we have } Ax = 0 \Leftrightarrow \begin{bmatrix} 2 & -3 & 1 \\ 1 & -2 & 1 \\ 1 & -3 & 2 \end{bmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \rightarrow \alpha = \beta = \gamma$$

$$x = \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \forall \alpha \in \mathbb{R}^* \quad / \text{ 1Mark}$$

Therefore $\lambda = 0$ is an eigenvalue of a matrix A and $x = \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \forall \alpha \in \mathbb{R}^*$ is an **eigenvector** or **characteristic**. / **1Mark**

$$\text{For } \lambda = 1, \text{ we have } (A - I)x = 0 \Leftrightarrow \begin{bmatrix} 1 & -3 & 1 \\ 1 & -3 & 1 \\ 1 & -3 & 1 \end{bmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \begin{pmatrix} \alpha - 3\beta + \gamma \\ \alpha - 3\beta + \gamma \\ \alpha - 3\beta + \gamma \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \rightarrow \alpha - 3\beta + \gamma = 0 \rightarrow \alpha = 3\beta - \gamma, \forall \beta, \gamma \in \mathbb{R}, \text{ not zero together.}$$

$$x = \begin{pmatrix} 3\beta - \gamma \\ \beta \\ \gamma \end{pmatrix} = \beta \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \gamma \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \quad / \text{ 1Mark}$$

Therefore $\lambda = 1$ is an eigenvalue of a matrix A and $x = \beta \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}, \forall \beta \in \mathbb{R}^*$ and $\gamma \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \forall \gamma \in \mathbb{R}^*$ are **eigenvectors** or **characteristic** belonging $\lambda = 1$ / **1Mark**

$$x = \gamma \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \forall \gamma \in \mathbb{R}^* \text{ are } \text{eigenvectors or characteristic belonging } \lambda = 1 \quad / \text{ 1Mark}$$